

Assignment 3

Bug Fix

Bug 1

```
for (let i = 0; i <= customers.length; i++) {  
  console.log(customers[i].name);  
}
```

```
Anita  
Kiran  
Meena  
Suresh  
C:\Training\Web\AlumulaRohithReddy-Week2-Assignment\Bugfix\main.js:11  
  console.log(customers[i].name);
```

Bug is index exceeding array index values, ($i < \text{customers.length}$) correct one.

After Fixing

```
Ravi  
Anita  
Kiran  
Meena  
Suresh
```

Bug 2

```
const activeCustomers = customers.filter((c) => {  
  c.active === true;  
});  
console.log(activeCustomers)
```

```
[]
```

Bug is return keyword missing

After fixing

```
[
  {
    id: 1,
    name: 'Ravi',
    age: 32,
    policy: 'Health',
    premium: 4800,
    active: true
  },
  {
    id: 2,
    name: 'Anita',
    age: 51,
    policy: 'Life',
    premium: 12000,
    active: true
  },
  {
    id: 4,
    name: 'Meena',
    age: 45,
    policy: 'Health',
    premium: 6000,
    active: true
  }
]
```

Bug 3

```
const updatedPremiums = customers.map((c) => {
  if (c.age >= 50) {
    c.premium = c.premium * 1.1;
  }
});
console.log(updatedPremiums)
```

```
[ undefined, undefined, undefined, undefined, undefined ]
```

Bug is return the object.

After Fixing

```
[
  {
    id: 1,
    name: 'Ravi',
    age: 32,
    policy: 'Health',
    premium: 4800,
    active: true
  },
  {
    id: 2,
    name: 'Anita',
    age: 51,
    policy: 'Life',
    premium: 13200.000000000002,
    active: true
  },
  {
    id: 3,
    name: 'Kiran',
    age: 28,
    policy: 'Vehicle',
    premium: 3500,
    active: false
  },
  {
    id: 4,
    name: 'Meena',
    age: 45,
    policy: 'Health',
    premium: 6000,
    active: true
  },
  {
    id: 5,
    name: 'Suresh',
    age: 60,
    age: 60,
    policy: 'Life',
    premium: 19800,
    active: false
  }
]
```

Bug 4

```
const totalPremium = customers.reduce((total, c) => {
  total + c.premium;
}, 0);
console.log(totalPremium)
```

undefined

Bug is return keyword missing

After Fixing

44300

Bug 5

```
console.log("Customer ${customers[0].name} has policy ${customers[0].policy}");
```

Customer \${customers[0].name} has policy\${customers[0].policy}

Bug is wrong quotes used.

After Fixing

Customer Ravi has policy Health

Bug 6

```
const policyCount = customers.reduce((count, c) => {  
  count.policy = (count.policy || 0) + 1;  
  return count;  
}, {});  
console.log(policyCount)
```

```
{ policy: 5 }
```

Bug is incorrect assignment of values in object count.

After Fixing

```
{ Health: 2, Life: 2, Vehicle: 1 }
```

Bug 7

```
const customersWithRisk = customers.map((c) => {  
  let riskLevel;  
  if (c.age < 35) riskLevel = "Low";  
  else if (c.age <= 50) riskLevel = "Medium";  
  else riskLevel = "High";  
  return { ...c, riskLevel };  
});  
console.log(customersWithRisk)
```

```
[  
  {  
    id: 1,  
    name: 'Ravi',  
    age: 32,  
    policy: 'Health',  
    premium: 4800,  
    active: true,  
    riskLevel: 'Medium'  
  },  
  {  
    id: 2,  
    name: 'Anita',  
    age: 51,  
    policy: 'Life',  
    premium: 12000,  
    active: true,  
    riskLevel: 'High'  
  },  
  {  
    id: 3,  
    name: 'Kiran',  
    age: 28,  
    policy: 'Vehicle',  
    premium: 3500,  
    active: false,  
    riskLevel: 'Medium'  
  },  
  {  
    id: 4,  
    name: 'Meena',  
    age: 45,  
    policy: 'Health',  
    premium: 6000,  
    active: true,  
    riskLevel: 'Medium'  
  },  
  {  
    id: 5,  
    name: 'Suresh',  
    age: 60,  
    policy: 'Life',  
    premium: 18000,  
    active: false,  
    riskLevel: 'High'  
  }  
]
```

Bug is else if condition missed

After Fixing

```
[
  {
    id: 1,
    name: 'Ravi',
    age: 32,
    policy: 'Health',
    premium: 4800,
    active: true,
    riskLevel: 'Low'
  },
  {
    id: 2,
    name: 'Anita',
    age: 51,
    policy: 'Life',
    premium: 12000,
    active: true,
    riskLevel: 'High'
  },
  {
    id: 3,
    name: 'Kiran',
    age: 28,
    policy: 'Vehicle',
    premium: 3500,
    active: false,
    riskLevel: 'Low'
  },
  {
    id: 4,
    name: 'Meena',
    age: 45,
    policy: 'Health',
    premium: 6000,
    active: true,
    riskLevel: 'Medium'
  },
  {
    id: 5,
    name: 'Suresh',
    name: 'Suresh',
    name: 'Suresh',
    age: 60,
    policy: 'Life',
    premium: 18000,
    active: false,
    riskLevel: 'High'
  }
]
```

Bug 8

```
let active = 0, inactive = 0;
for (const c in customers) {
  if (c.active) active++;
  else inactive++;
}
console.log(active, inactive)
```

0 5

Bug is use of for..in instead of for..of

After Fixing

3 2

Bug 9

```
const getLifeCustomers = () =>
  customers.filter((c) => c.policy === "Life").map((c) =>
    c.name);
```

```
console.log(getLifeCustomers())
```

Bug is return keyword needed if {} brackets are used.

After Fixing

```
const getLifeCustomers = () =>{  
  return customers.filter((c) => c.policy === "Life").map((c) =>  
    c.name); }  
console.log(getLifeCustomers())
```

Bug 10

```
const sortedCustomers = customers.sort((a, b) => b.premium - a.premium);  
console.log(customers)
```

```
[  
  {  
    id: 5,  
    name: 'Suresh',  
    age: 60,  
    policy: 'Life',  
    premium: 18000,  
    active: false  
  },  
  {  
    id: 2,  
    name: 'Anita',  
    age: 51,  
    policy: 'Life',  
    premium: 12000,  
    active: true  
  },  
  {  
    id: 4,  
    name: 'Meena',  
    age: 45,  
    policy: 'Health',  
    premium: 6000,  
    active: true  
  },  
  {  
    id: 1,  
    name: 'Ravi',  
    age: 32,  
    policy: 'Health',  
    premium: 4800,  
    active: true  
  },  
  {  
    id: 3,  
    name: 'Kiran',  
    age: 28,  
    policy: 'Vehicle',  
    premium: 3500,  
    active: false  
  }  
]
```

Bug is that the original array order will also change when we perform sort on it so make a copy and sort

After Fixing

```
[
  {
    id: 1,
    name: 'Ravi',
    age: 32,
    policy: 'Health',
    premium: 4800,
    active: true
  },
  {
    id: 2,
    name: 'Anita',
    age: 51,
    policy: 'Life',
    premium: 12000,
    active: true
  },
  {
    id: 3,
    name: 'Kiran',
    age: 28,
    policy: 'Vehicle',
    premium: 3500,
    active: false
  },
  {
    id: 4,
    name: 'Meena',
    age: 45,
    policy: 'Health',
    premium: 6000,
    active: true
  },
  {
    id: 5,
    name: 'Suresh',
    age: 60,
    policy: 'Life',
    premium: 18000,
    active: false
  }
]
```